Tutorial 1 : Introduction to optimization and refresher course

Exercise 1 (Differentiation of some functions)**.** Compute the gradient and Hessian of the following functions:

1. $f : \mathbb{R}^d \to \mathbb{R} : x \mapsto \|\mathbf{A}x - b\|_2^2$ ($\mathbf{A}$ and b are constant matrix and vector).
2. $f : \mathbb{R}^d \to \mathbb{R} : x \mapsto x^\top \mathbf{A}x - b^\top x + c$ ($\mathbf{A}, b, c$ are constant matrix, vector and scalar).
3. $f : \mathbb{R}^d \to \mathbb{R} : x \mapsto \|x\|_2^a$ (where $a > 2$).
4. $g : \mathbb{R} \to \mathbb{R} : t \mapsto f(x + t(y - x))$ (x, y are two fixed vectors, f is a fixed C^2 function). Express the gradient and the Hessian matrix of g by those of f.

Exercise 2 (Differentiable but not C^1)**.** Consider the function:

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

1. Is f differentiable?
2. Is f continuously differentiable?
3. Based on this function, can you construct a function f such that f is twice differentiable but not C^2?

Exercise 3 (Necessary conditions of optimal solution revisited)**.** If f is only differentiable and not C^1, is it still necessary that $\nabla f(x^\star) = 0$ for any local solution $x^\star$?

Exercise 4 (Properties of derivatives and gradient)**.** Given two differentiable functions $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$, we have:

$$\begin{aligned} \nabla(f+g)(x) &= \nabla f(x) + \nabla g(x) \\ \nabla(\alpha f)(x) &= \alpha \nabla f(x), \forall \alpha > 0 \\ \nabla(f \cdot g)(x) &= g(x)\nabla f(x) + f(x)\nabla f(x) \\ \nabla\left(\frac{f}{g}\right) &= \frac{g(x)\nabla f(x) - f(x)\nabla g(x)}{g(x)^2}, \quad \text{assuming that } g(x) > 0. \end{aligned} \tag{1}$$

Exercise 5 (Chain rule)**.** Given two differentiable functions $f : \mathbb{R}^k \to \mathbb{R}^\ell$ and $g : \mathbb{R}^d \to \mathbb{R}^k$, prove that the composition $f \circ g : \mathbb{R}^d \to \mathbb{R}^\ell$ is also differentiable and its Jacobian matrix is given by:

$$J_{f \circ g}(x) = J_f(g(x)) J_g(x).$$

Exercise 6 (Two Taylor formulations)**.** Given a C^1 (resp. C^2) function $f : \mathbb{R}^d \to \mathbb{R}$, we have:

$$\begin{aligned} f(y) &= f(x) + \int_0^1 \nabla f(x + t(y-x))^\top (y-x)\, dt && ,\forall x, y \in \mathbb{R}^d \\ \text{(resp.)} f(y) &= f(x) + (y-x)^\top \nabla f(x) + \frac{1}{2}(y-x)^\top \nabla^2 f(x)(y-x) + R_2(x-y) && ,\forall x, y \in \mathbb{R}^d, \end{aligned} \tag{2}$$

where $R_2(x-y)$ is a reminder satisfying $\lim_{y \to x} \frac{R_2(x-y)}{\|y-x\|^2} = 0$.
Hint: you might need to use the fundamental theorem of calculus, i.e., if $f : \mathbb{R} \to \mathbb{R}$ is differentiable, we have:

$$f(b) = f(a) + \int_a^b f'(t) dt.$$

Tutorial 2 : Convex sets and functions

Exercise 1 (Unproven proposition)**.** If A is a convex set, then any convex combination of finite points of A is an element of A, i.e., for $x_1, \ldots, x_k \in A$, $\alpha_1, \ldots, \alpha_k \geq 0, \sum_i \alpha_i = 1$, we have:

$$\alpha_1 x_1 + \ldots + \alpha_k x_k \in A.$$

Exercise 2 (Convex sets)**.** Decide whether the following sets are convex and justify your choice:

1. A slab, i.e., a set of the form: $\{x \in \mathbb{R}^n \mid \alpha \leq x^\top a \leq \beta\}$ (α, β, a are fixed scalars and vector).
2. A rectangle, i.e., a set of the form: $\{x \in \mathbb{R}^d \mid \alpha_i \leq x_i \leq \beta_i\}$.
3. A wedge, i.e., $\{x \in \mathbb{R}^d \mid a_1^\top x \leq b_1, a_2^\top x \leq b_2\}$.
4. The set of points closer to a given point than a given set, i.e.,
$$\{x \in \mathbb{R}^d \mid \|x - x_0\|_2 \leq \|x - y\|_2, \forall y \in S\}$$
5. The set of points closer to one set than another, i.e.,
$$\{x \in \mathbb{R}^d \mid \mathrm{dist}(x, S) \leq \mathrm{dist}(x, T)\}$$
where $\mathrm{dist}(x, S) = \inf_{y \in S} \|x - y\|_2$.
6. The set $\{x \in \mathbb{R}^d \mid x + S_2 \subseteq S_1\}$ where $S_1, S_2 \subseteq \mathbb{R}^d$ and S_1 is convex.
7. The set of points whose distance to a does not exceed a fixed fraction θ of the distance to b, i.e., $\{x \in \mathbb{R}^d \mid \|x - a\|_2 \leq \theta \|x - b\|_2\}$. You can assume that $a \neq b$ and $\theta \in [0, 1]$.

Exercise 3 (Operations preserving convexity)**.** Prove that the following operations preserve the convexity:

1. If f is convex, then αf is also convex ($\alpha > 0$).
2. If f and g are convex, then $f + g$ is also convex.
3. If $f_i, i = 1, \ldots, n$ are convex and $\alpha_i \geq 0, \forall i = 1, \ldots, n$, then $\sum_i \alpha_i f_i$ is also convex.
4. If f, g are convex, then $\max(f, g)$ is convex.
5. If f is convex, then $g(x) = f(Ax + b)$ is also convex.

6. if $f : \mathbb{R}^d \to \mathbb{R}$ is convex and $g : \mathbb{R} \to \mathbb{R}$ is non-decreasing and convex, then $g \circ f$ is convex.

Proof. To be done. □

Exercise 4 (Sublevel sets of convex functions)**.** Given a function $f : \mathbb{R}^d \to \mathbb{R}$, the α, the sublevel sets of f is given by:

$$\{x \in \mathbb{R}^d \mid f(x) \leq c\}.$$

1. Prove that the sublevel set of a convex function is convex.
2. Is the converse correct, i.e., if all the sublevels of f are convex, is f necessarily a convex function?
3. Applying this result to show that the ellipsoid, i.e., $\{x \in \mathbb{R}^d \mid (x-y)^\top \mathbf{A}(x-y) \leq c\}$ for some $\mathbf{A} \in \mathbb{R}^{d\times d}, \mathbf{A} \succeq 0, y \in \mathbb{R}^d, c \in \mathbb{R}$.

Exercise 5 (Optimal condition for constrained convex optimization)**.** Consider a constrained optimization problem given by:

$$\underset{x\in\mathcal{F}}{\text{Minimize}} \quad f(x)$$

where f is C^1 and convex, $\mathcal{F}$ is a convex set. A point x is a global solution of $f(x)$ if and only if $\nabla f(x)^\top (y-x) \geq 0, \forall y \in \mathcal{F}$.

Exercise 6 (Projection onto a convex set)**.** In this exercise, we will prove for any non-empty closed convex set $S \subseteq \mathbb{R}^d, x \in \mathbb{R}^d$, the set of projector of x onto S is a singleton, i.e., $\{y \in S \mid \|x-y\|_2 = \text{dist}(x, S)\}$. To do so, answer the following questions:

1. Prove that the set of projectors of x onto S is non-empty.
2. Let y be a projector of x onto S, prove that:

$$(y-x)^\top (z-y) \geq 0, \forall z \in S.$$

   Hint: Use the previous exercise.

3. Prove that the set of projectors has at most one element by contradiction.

Proof. Proof is left as exercise. □

Exercise 7 (Seperating plane of disjoint convex sets)**.** In this exercise, we will prove the following results: For any two non-empty, convex sets $C, D \subseteq \mathbb{R}^d$ such that $C \cap D = \emptyset$, there exists a separating hyperplane, i.e., a vector $a \in \mathbb{R}^d, b \in \mathbb{R}$ satisfying:

$$x \in C \implies x^\top a \leq b,$$
$$x \in D \implies x^\top a \geq b.$$

Prove a simpler version of this result where we assume that there exists $c \in C, d \in D$ such that $\|x-y\|_2 = \text{dist}(C, D)$ by following these steps:

1. Define $a = d - c$, $b = \frac{\|d\|^2 - \|c\|^2}{2}$. Prove that:

$$f(x) = x^\top a - b = (d-c)^\top \left(x - \frac{1}{2}(d+c)\right) = \frac{1}{2}\|d-c\|^2 + (d-c)^\top (x-d).$$

2. Prove that for $x \in D$, we have: $(d-c)^\top (x-d) \geq 0$.
3. Conclude that for $x \in D$, $x^\top a \geq b$. Conclude the proof by making the same argument can be used to prove that $x \in C \implies x^\top a \leq b$.

Tutorial 3 : Gradient descent and theoretical properties

Exercise 1 (Gradient descent update)**.** Given a sequence of pair $(x_i, y_i), y_i \in \{\pm 1\}$, we consider the following optimization problem (also known as *logistic regression*).

$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log(1 + \exp(-y_i x_i^\top \theta)). \tag{1}$$

Prove that the update of gradient descent for $\mathcal{L}$ is given by:

$$\theta_{k+1} = \theta_k - \frac{\alpha}{n}\sum_{i=1}^{n} \frac{-y_i}{1 + \exp(y_i x_i^\top \theta_k)} x_i.$$

Exercise 2 (Unproven proposition 1)**.** If f is L-smooth, then for all $x, y \in \mathbb{R}^d$, we have:

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|x - y\|_2^2.$$

Exercise 3 (Unproven proposition 2)**.** Consider $f : \mathbb{R}^d \to \mathbb{R}$ a μ-strongly convex function, we have:

1. $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - \frac{1}{2}\mu t(1-t)\|x - y\|^2, \forall x, y \in \mathbb{R}^d, \forall t \in [0, 1]$.
2. If f is C^1, then $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2}\|x - y\|^2, \forall x, y \in \mathbb{R}^d$.
3. $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \mu\|x - y\|^2, \forall x, y \in \mathbb{R}^d$.
4. If f is C^2, then $\nabla^2 f(x) \succeq \mu \mathbf{I}$ (i.e., $\nabla^2 f(x) - \mu \mathbf{I}$ est positive semidefinite).

Exercise 4 (Gradient descent on quadratic optimization)**.** Consider a simple quadratic optimization of the form:

$$f(x) = \frac{1}{2} x^\top \mathbf{A} x$$

where $\mathbf{A} \in \mathbb{S}^{d \times d}$ is a symmetric matrix. Remind that if $\mathbf{A} \in \mathbb{S}^{d \times d}$ is a symmetric matrix, there exists an orthogonal $\mathbf{Q} \in \mathbb{R}^{d \times d}$ and a diagonal matrix $\mathbf{D} \in \mathbb{R}^{d \times d}$ such that $\mathbf{A} = \mathbf{Q}^\top \mathbf{D} \mathbf{Q}$. Answer the following question:

1. Prove that f is convex if and only if $\mathbf{D}$ has nonnegative entries. Deduce that f is μ-strongly convex if and only if all the coefficients in the diagonal of $\mathbf{D}$ are at least μ.
2. If f is convex, prove that $f^\star = \min_x f(x) = 0$.

3. If f is μ-strongly convex, what can we deduce about the convergence speed $f(x_k)$ to 0 if x_k is generated by gradient descent (with a proper choice of the learning rate).

4. If f is only convex, can we say the same thing about the convergence speed of $f(x_k)$ obtained by gradient descent (with a proper choice of the learning rate).

Exercise 5 (Armijo and Wolfe conditions)**.** Armijo (A) and Wolfe (W) conditions provides criteria to choose the step-size $\alpha_k > 0$ when minimizing a C^1 function f. Assume that at the kth iteration, one considers update the $(k+1)$th iterate as:

$$x_{k+1} = x_k + \alpha p_k,$$

these two conditions requires:

$$f(x_{k+1}) \leq f(x_k) + \alpha_k c_1 \nabla f(x_k)^\top p_k, \qquad c_1 \in (0,1) \tag{A}$$

$$\nabla f(x_{k+1})^\top p_k \geq c_2 \nabla f(x_k)^\top p_k, \qquad c_2 \in (c_1, 1) \tag{W}$$

Our goal is to prove that if f is C^1 and bounded below, $p_k^\top \nabla f(x_k) < 0$, then there exists $\alpha > 0$ satisfying (A) and (W). Consider two functions:

$$\varphi(\alpha) = f(x_k + \alpha p_k) \quad , \quad \ell(\alpha) = f(x_k) + \alpha c_1 \nabla f(x_k)^\top p_k.$$

1. Define $g(\alpha) = \varphi(\alpha) - \ell(\alpha)$. Prove that there is an interval $(0, \bar{\alpha})$ such that $g(\alpha) > 0, \forall \alpha \in (0, \bar{\alpha})$ and $g(0) = g(\bar{\alpha}) = 0$.

2. By mean value theorem, prove that there exists $\tilde{\alpha}$ such that:

$$\frac{\varphi(\bar{\alpha}) - \varphi(0)}{\bar{\alpha}} = \varphi'(\tilde{\alpha}).$$

3. Conclude that $\tilde{\alpha}$ satisfies both (A) and (W).

Exercise 6 (More about line search)**.** Consider the line search conditions (A) and (W).

1. Show that if $0 < c_2 < c_1 < 1$, there may not exists α_k satisfying both (A) and (W).

2. Prove that if α_k satisfies (W) for some $c_2 \in (0,1)$ and $\nabla f(x_k)^\top p_k < 0$, then the following equation (a.k.a curvature condition) is satisfied:

$$(x_{k+1} - x_k)^\top (\nabla f(x_{k+1}) - \nabla f(x_k)) > 0.$$

Exercise 7 (Linear convergence of iterates)**.** Consider a μ-strongly convex, L-smooth function f. Prove that the iterates generated by gradient descent with step-size $1/L$ satisfying that:

$$\|x_k - x^\star\|_2^2 = O\left(\left(1 - \frac{\mu}{L}\right)^k\right).$$

where $x^\star$ is the unique global optimal solution of f.

Tutorial 4 : Gradient descent and theoretical properties

Exercise 1 (Unproved proposition)**.** Given a C^2 function f. Prove that the following statements are equivalent:

1. f is L-smooth.
2. For any $x \in \mathbb{R}^d$, $\|\nabla^2 f\|_{\text{op}} \leq L$ where $\|\cdot\|_{\text{op}}$ of a symmetric matrix is defined by:
$$\|\mathbf{A}\| = \max_{i=1,\dots,n} |\lambda_i(\mathbf{A})| \qquad (\lambda_i \text{ is the } i\text{th eigenvalue of } \mathbf{A}).$$
3. $-L\mathbf{I} \preceq \nabla^2 f(x) \preceq L\mathbf{I}, \forall x \in \mathbb{R}^n$.

Exercise 2 (Chebyshev polynomials)**.** For a given $n \in \mathbb{N}$, the Chebyshev polynomial of the first kind is given as the polynomial T_n such that:

$$T_n(\cos\theta) = \cos(n\theta).$$

Answer the following questions:

1. Prove that T_n can be defined alternatively as:
$$\begin{aligned} T_0(x) &= 1, \\ T_1(x) &= x, \\ T_{n+1}(x) &= 2xT_n(x) - T_{n-1}(x). \end{aligned}$$
2. Consider $f(x) = \frac{1}{2^{n-1}} T_n(x)$. Prove that the leading coefficient of $f(x)$ is 1.
3. Prove that $\max_{x\in[-1,1]} |f(x)| = \frac{1}{2^{n-1}}$.
4. Prove that for any polynomial P of degree n whose leading term is 1, $\max_{x\in[-1,1]} |f(x)| \geq \frac{1}{2^{n-1}}$. Hint: Assume a "bad" polynomial ω exists: Consider the function $g = f - \omega$. What is the degree of g? And what is the lower-bound of the number of roots of g?

Exercise 3 (Worst quadratic function)**.** The function that we used to prove the lower-bound of first-order methods is actually quadratic. To understand more about this function, consider a slightly more simple function:

$$f_k(x) = \frac{1}{2}\left[x_1^2 + \sum_{i=1}^{k-1}(x_i - x_{i+1})^2 + x_k^2\right] = \frac{1}{2}x^\top \mathbf{A}_k x.$$

Answer the following question:

1. Prove that $\mathbf{A}_k$ takes the following form:

$$\mathbf{A}_k = \left(\begin{array}{c|c} \begin{pmatrix} 2 & -1 & 0 & & & \\ -1 & 2 & -1 & & 0 & \\ 0 & -1 & 2 & & & \\ & \cdots & & & \cdots & \\ & 0 & & -1 & 2 & -1 \\ & & & 0 & -1 & 2 \end{pmatrix} & \mathbf{0}_{k\times(n-k)} \\ \hline \mathbf{0}_{(n-k)\times k} & \mathbf{0}_{(n-k)\times(n-k)} \end{array}\right)$$

2. Prove that $\mathbf{A}_k$ has $(n-k)$ zero eigenvalues and k positive eigenvalues given by:

$$2 - 2\cos\left(\frac{\pi j}{k+1}\right) = 4\sin^2\left(\frac{j\pi}{2(k+1)}\right), \quad j = 1, \ldots, k$$

   Hint: You might want to use the fact that the sequence $x_0 = 1, x_1 = 2\cos(\theta), x_{k+1} = 2\cos(\theta)x_k - x_{k-1}$, then $x_k = \frac{\sin((k+1)\theta)}{\sin(\theta)}$.

3. Is the function μ-strongly convex ($\mu > 0$)? If yes, for which value of μ.

4. Is the function L-smooth? If yes, for which value of L.

5. What are the minimizer of the function $f_k, k = 1, \ldots, n$?

6. For a function f that is μ-strongly convex and L-smooth, remind that we have linear convergence of f, i.e.,

$$f(x_k) - f(x^\star) \leq O\left(\left(1 - \frac{\mu}{L}\right)^k\right)$$

   Does this contradict the lower-bound that we establish in the lecture?

Tutorial 5 : Stochastic gradient descent and theoretical properties

Exercise 1 (Optimization problems satisfy the hypothesis)**.** Prove that the stochastic gradient descent of the following cases satisfy our hypothesis on g_k, i.e., $\mathbb{E}[g_k] = \nabla\mathcal{L}$ and $\texttt{Var}[g_k] \leq \sigma^2$ for some fixed constant $\sigma > 0$.

1. Gradient pertubed by Gaussian noise, i.e., $g_k(\theta) = \nabla\mathcal{L}(\theta) + \epsilon_k$ and $\epsilon_k \sim \mathcal{N}(0, \sigma^2\mathbf{I})$.
2. We consider the optimization problem of a linear regression problem of the form:

$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \underbrace{\log(1 + \exp(-y_i x_i^\top \theta))}_{\ell_i(\theta)}. \tag{1}$$

   and $g_i(x) = \nabla\ell_i(x), i \sim \mathcal{U}(\{1, \ldots, n\})$ (i is sampled uniformly from the set $\{1, \ldots, n\}$).

Exercise 2 (Reduce the variant in stochastic gradient descent)**.** We saw that the variance of $\sigma^2 = \texttt{Var}[\|g_k\|^2]$ plays an important role in the analysis of stochastic gradient descent. The smaller is σ, the better the bound is. In reality, practitioners employ many techniques to reduce σ. Minibatching is one of them. Indeed, consider an optimization problem of the form:

$$\underset{\theta\in\mathbb{R}^d}{\text{Minimize}} \quad \mathcal{L}(\theta) = \sum_{i=1}^{n} \ell_i(\theta).$$

Instead of taking $g_k = \nabla\ell_i(\theta_k)$, minibatching takes:

$$\tilde{g}_k^2 = \frac{1}{|S|}\sum_{i\in S} \nabla\ell_i(\theta_k).$$

where $S \subseteq \{1, \ldots, n\}$ is uniformly sampled from $\{1, \ldots, n\}$ with replacement. Prove that:

$$\texttt{Var}[\|\tilde{g}_k\|^2] = \frac{1}{|S|}\texttt{Var}[\|g_k\|^2].$$

Exercise 3 (Stochastic gradient descent with convex and L-smooth functions)**.** Consider the stochastic gradient descent when optimizing a convex and L-smooth function f. Assume that the stochastic gradient g_k satisfy the hypothesis as in the lecture. We want to prove that if the step size $\alpha_k = \alpha < \frac{1}{2L}$, then:

$$\mathbb{E}[f(\bar{x}^T) - \inf f] \leq \frac{\|x_0 - x^\star\|^2}{T\alpha} + \sigma^2\alpha.$$

where $\bar{x}^T = \frac{1}{T}\sum_{i=0}^{T-1} x_i$. Follow these steps:

1. Prove that:

$$\mathbb{E}[\|x_{k+1} - x^\star\|^2 | x_k] = \|x_k - x^\star\|^2 - 2\alpha \nabla f(x_k)^\top (x_k - x^\star) + \alpha^2(\|\nabla f(x_k)\|^2 + \sigma^2)$$

2. Prove that:

$$\mathbb{E}[\|x_{k+1} - x^\star\|^2 | x_k] \leq \|x_k - x^\star\|^2 + 2\alpha(1 - L\alpha)(f(x_k) - f(x^\star)) + \alpha^2\sigma^2$$

3. Conclude.

4. Optimizing α to get the best bound for the timestep K.

Exercise 4 (Choice of step-size)**.** Consider the same setting as the previous exercise. However, this time, we will choose α_k differently and $\alpha_k < \frac{1}{2L}, \forall k \in \mathbb{N}$. In that case, the previous result becomes:

$$\mathbb{E}[f(\bar{x}^T) - \inf f] \leq \frac{\|x_0 - x^\star\|^2}{\sum_i \alpha_i} + \sigma^2 \frac{\sum_i \alpha_k^2}{\sum_i \alpha_i}.$$

where $\bar{x}^T = \frac{\sum_{i=0}^{T-1} x_i \alpha_i}{\sum_i \alpha_i}$ (weighted average).

1. Using the same schema, reprove the above theoretical guarantee.

2. The Robbins-Monro step-sizes are sequences of $\{\alpha_k\}_{k\in\mathbb{N}}$ satisfies:

   (a) $\sum_k \alpha_k^2 < \infty$.

   (b) $\sum_k \alpha_k$ is not bounded.

   Prove that this choice allows $\mathbb{E}[f(\bar{x}^T) - \inf f] \to 0$.

3. For which value of β that $\alpha_k = k^{-\beta}$ satisfies Robbins Monro criteria.

4. Which values of β yields the best asymptotic convergence rate.

Exercise 5 (Subgaussian distributions)**.** In mathematical analysis of stochastic optimization algorithms, many works assume that the noise ϵ_k belongs to certain families of distribution. Subgaussian, a generalization of Gaussian distribution is among the most popular. In this exercise, we take a look at this family. Prove the following equivalent definition of a subgaussian distribution: Let X be a random variable:

1. There exists $K_1 > 0$ such that:

$$\mathbb{P}(|X| \geq t) \leq 2\exp\left(-\frac{t^2}{K_1^2}\right), \forall t \geq 0.$$

2. There exists $K_2 > 0$ such that:

$$\mathbb{E}[|X|^p]^{\frac{1}{p}} \leq K_2\sqrt{p}, \forall p > 1.$$

3. There exists $K_3 > 0$ such that:

$$\mathbb{E}\exp\left(\frac{X^2}{K_3}\right) \leq 2.$$

Tutorial 6 : Adaptive methods - Adagrad and variants

In this tutorial, we consider a simpler version of Adagrad: Adagrad-Norm. It is defined as follows:

$$\begin{aligned} v_{k+1} &= v_k + \|\nabla f(x_k)\|^2, \qquad v_0 = 0 \\ x_{k+1} &= x_k - \alpha \frac{1}{\sqrt{v_{k+1}}} \nabla f(x_k). \end{aligned} \tag{AdaGrad-Norm}$$

Exercise 1 (Warm-up)**.** Answer the following question:

1. Is (AdaGrad-Norm) a first-order method in the Nesterov sense, i.e.:

$$x_{k+1} \in x_0 + \mathtt{Lin}\left\{\nabla f(x_k), \nabla f(x_{k-1}), \ldots, \nabla f(x_0)\right\}.$$

2. Consider applying (AdaGrad-Norm) to minimize the absolute value function, i.e., $f(x) = |x|$. Prove that if $x_i \neq 0, \forall i = 0, \ldots, k$, we have:

$$x_{k+1} = x_k - \alpha \frac{\mathtt{sign}(x_k)}{\sqrt{k+1}}.$$

3. Prove that $\lim_{k\to\infty} x_k = 0$ (Warning: this question might be complicated!!!).

Exercise 2 (First properties)**.** Consider $\{x_k\}_{k\in\mathbb{N}}$ the iterates of (AdaGrad-Norm), prove that:

1. For any $T \geq 0$ and $x \in \mathbb{R}^d$, we have:

$$\sum_{k=0}^{T-1} \langle \nabla f(x_k), x_k - x\rangle \leq \left(\frac{\max_{1\leq t\leq T-1} \|x_t - x\|^2}{2\alpha} + \alpha \right) \sqrt{v_T}.$$

   Hint: you might want to use the inequalities:

$$\sum_{k=0}^{T-1} \frac{\|\nabla f(x_k)\|^2}{\sqrt{v_{k+1}}} \leq \sqrt{v_T}$$

2. Assume that $\max_{1\leq t\leq T-1} \|x_t - x\|^2 \leq R^2$, optimizing α to get the following inequalities:

$$\sum_{k=0}^{T-1} \langle \nabla f(x_k), x_k - x\rangle \leq R\sqrt{2v_T}.$$

Exercise 3 (Convex optimization problem)**.** Consider $\{x_k\}_{k\in\mathbb{N}}$ the iterates of (AdaGrad-Norm) optimizing a convex, L-Lipschitz function f with at least one minimizer $x^\star$. Prove that:

1. For any $T \geq 0$ and $x \in \mathbb{R}^d$, we have:

$$\min_{k=0,\dots,T-1} f(x_k) - f^\star \leq \left(\frac{\max_{0\leq t\leq T-1} \|x_t - x\|^2}{2\alpha} + \alpha \right) \frac{L}{\sqrt{T}}.$$

2. Assume that $\max_{1\leq t\leq T-1} \|x_t - x\|^2 \leq R$, optimizing α to get the following inequalities:

$$\min_{k=0,\dots,T-1} f(x_k) - f^\star \leq RL\sqrt{\frac{2}{T}}.$$

Exercise 4 (Smooth convex optimization problem)**.** Consider $\{x_k\}_{k\in\mathbb{N}}$ the iterates of (AdaGrad-Norm) optimizing a convex, L-smooth function f with at least one minimizer $x^\star$. Prove that:

1. For any $T \geq 0$ and $x \in \mathbb{R}^d$, we have:

$$\min_{k=0,\dots,T-1} f(x_k) - f^\star \leq \left(\frac{\max_{0\leq t\leq T-1} \|x_t - x\|^2}{2\alpha} + \alpha \right)^2 \frac{2L}{T}.$$

2. Assume that $\max_{1\leq t\leq T-1} \|x_t - x\|^2 \leq R$, optimizing α to get the following inequalities:

$$\min_{k=0,\dots,T-1} f(x_k) - f^\star \leq \frac{4R^2L}{T}.$$

Tutorial 7 : Second-order methods - Newton algorithm

Exercise 1 (Newton method for quadratic functions)**.** Consider a quadratic function $f(x) = \frac{1}{2}x^\top \mathbf{A}x + b^\top x + c$, where $\mathbf{A} \succ 0$.

1. What are the minimizers of f?
2. What is the pure Newton step for the function f?
3. How many steps does it take for the Newton method to reach the minimizers of f?

Exercise 2 (Divergence of pure Newton steps)**.** If one only performs pure Newton steps, the iterates might diverge, even if the function is convex. Consider:

$$f(x) = -\log x + x$$

Answer the following questions:

1. Is the function f convex? Is the function f strongly convex? What is its minimizer?
2. Write the pure Newton step for the function f.
3. What is the set of initialization x_0 so that pure Newton steps can converge to its minimizer? What is the convergence speed? Can we apply the theoretical results proved in the lecture?
4. Bonus: Answer the same questions with the function $f(x) = \log(\exp(x) + \exp(-x))$.

Exercise 3 (BFGS algorithm)**.** The BFGS algorithm is a quasi-Newton method, i.e., it shares a similar model to the Newton algorithm. Indeed, at the kth iteration, BFGS constructs a quadratic surrogate of f of the form:

$$m_k(p) = f(x_k) + \nabla f(x_k)^\top p + \frac{1}{2}p^\top \mathbf{B}_k p.$$

where $\mathbf{B}_k \succ 0$. We will update x_{k+1} as:

$$\begin{aligned} x_{k+1} &= x_k - \alpha_k \mathbf{B}_k^{-1} \nabla f(x_k). \\ \mathbf{B}_{k+1} &= (\mathbf{I} - \rho_k y_k s_k^\top)\mathbf{B}_k(\mathbf{I} - \rho_k s_k y_k^\top) + \rho_k y_k y_k^\top \\ s_k &= x_{k+1} - x_k, \qquad y_k = \nabla f(x_{k+1}) - \nabla f(x_k), \qquad \rho_k = \frac{1}{y_k^\top s_k}. \end{aligned}$$

Answer the following questions:

1. Prove that if $\mathbf{B}_k \succ 0$, then $p_k = -\mathbf{B}_k^{-1}\nabla f(x_k)$ is a descent direction, i.e., $p_k^\top \nabla f(x_k) < 0$.
2. Assume the step-size α_k satisfies the Wolfe condition, i.e.,:
$$\nabla f(x_{k+1})^\top p_k \geq c_2 \nabla f(x_k)^\top p_k, \quad c_2 \in (0,1),$$
prove that $\rho_k > 0$.
3. Prove that if $\mathbf{B}_k \succ 0$, then $\mathbf{B}_{k+1} \succ 0$.

Exercise 4 (How to derive B_{k+1} in BFGS algorithm?)**.** In this exercise, we will prove that:

$$\mathbf{B}_{k+1} \in \operatorname*{argmin}_{\mathbf{B}\succ 0} \{\|\mathbf{B} - \mathbf{B}_k\|_{\mathbf{W}} \mid \mathbf{B}s_k = y_k\}$$

where $\mathbf{W}$ is any positive definite matrix satisfying $\mathbf{W}y_k = s_k$ and $\|\mathbf{A}\|_{\mathbf{W}} = \|\mathbf{W}^{\frac{1}{2}}\mathbf{A}\mathbf{W}^{\frac{1}{2}}\|_F$. Answer the following questions:

1. Let $\mathbf{W}^{1/2}\mathbf{B}_k\mathbf{W}^{1/2} = \hat{\mathbf{B}}_k$, $\mathbf{W}^{1/2}\mathbf{B}\mathbf{W}^{1/2} = \hat{\mathbf{B}}$, $\mathbf{W}^{1/2}y_k = \hat{y}_k, \mathbf{W}^{-1/2}s_k = \hat{s}_k$. Prove that our optimization problem can be reformulated as:
$$\arg\min\{\|\hat{\mathbf{B}} - \hat{\mathbf{B}}_k\|_F \mid \hat{\mathbf{B}} \succ 0, \hat{\mathbf{B}}\hat{\mathbf{s}}_k = \hat{\mathbf{s}}_k\}. \tag{1}$$
2. Let $u = \frac{\hat{s}_k}{\|\hat{s}_k\|} \in \mathbb{R}^d$ and $u_\perp \in \mathbb{R}^{d\times(d-1)}$ be any orthonormal complement of u (i.e., $(u \quad u_\perp) \in \mathbb{R}^{d\times d}$ is a unitary matrix). Prove the optimality condition for problem (1) can be written as
$$u_\perp^\top \hat{\mathbf{B}} u_\perp = u_\perp^\top \hat{\mathbf{B}}_k u_\perp.$$
3. Prove that the optimal solution of (1) is
$$\hat{\mathbf{B}} = uu^\top + (\mathbf{I} - uu^\top)\hat{\mathbf{B}}_k(\mathbf{I} - uu^\top).$$
4. Recover the original formula.

Exercise 5 (Fast update rule for BFGS)**.** In the BFGS algorithm, we only care about the product of $\mathbf{B}_k^{-1}$ with a vector. Therefore, it is better to update $\mathbf{B}_{k+1}^{-1}$ from $\mathbf{B}_k^{-1}$ directly. Prove that:

1. Sherman - Morrison – Woobury Formula: if $\mathbf{A} = \mathbf{B} + \mathbf{U}\mathbf{C}\mathbf{V}^\top$, then
$$\mathbf{A}^{-1} = \mathbf{B}^{-1} - \mathbf{B}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}^\top\mathbf{B}^{-1}\mathbf{U})^{-1}\mathbf{V}^\top\mathbf{B}^{-1}$$
2. Prove that:
$$\mathbf{B}_{k+1}^{-1} = \mathbf{B}_k^{-1} + \frac{1}{s_k^\top y_k} s_k s_k^\top - \frac{1}{y_k^\top \mathbf{B}_k^{-1} y_k}\mathbf{B}_k^{-1} y_k y_k^\top \mathbf{B}_k^{-1}.$$

Exercise 6 (Another version of BFGS)**.** In reality, the most implemented version of BFGS uses a slightly different update rule from the one described in the previous exercise. Indeed, they choose to update $\mathbf{H}_k$, the inverse of $\mathbf{B}_k$ as:

$$\mathbf{H}_{k+1} \in \operatorname*{argmin}_{\mathbf{H}\succ 0} \{\|\mathbf{H} - \mathbf{H}_k\|_{\mathbf{W}} \mid \mathbf{H}y_k = s_k\}$$

where $\mathbf{W}$ is any positive definite matrix satisfying $\mathbf{W}s_k = y_k$. The descent direction is then $\mathbf{H}_k\nabla f(x_k)$ (and not $\mathbf{B}_k^{-1}\nabla f(x_k)$). Can you derive the update rule of $\mathbf{H}_k$?

Tutorial 8 : Nonsmooth optimization - Subgradient Descent

Exercise 1 (Subgradient of functions)**.** Compute the subgradients of the following functions:

1. $\mathrm{ReLU}(x) = \max(x, 0)$.
2. ℓ_1 norm, i.e. $\|x\|_1 = \sum_{i=1}^d |x_i|$.
3. Generalize the previous question for decomposable functions, i.e., compute the subgradient $f(x) = \sum_{i=1}^d f_i(x_i)$ with those of $f_i, i = 1, \ldots, d$.
4. ℓ_2 norm, i.e., $\|x\|_2 = \sqrt{\sum_{i=1}^d x_i^2}$.
5. Total variation function, i.e., $\mathrm{TV}(x) = \sum_{i=1}^{d-1} |x_{i+1} - x_i|$.
6. $f(x) = \|\mathbf{A}_1 x + b\|_1 + \|x\|_2, \mathbf{A} \in \mathbb{R}^{m\times d}, b \in \mathbb{R}^m$.

Exercise 2 (Normal cone introduction)**.** A very important concept in optimization is the normal cone. More formally, a normal cone to a convex set A at a point $x \in A$ is defined as:

$$\mathcal{N}(x, A) := \{u \in \mathbb{R}^n \mid \langle v - x, u\rangle \leq 0, \forall v \in A\}.$$

Answer the following questions:

1. $\mathcal{N}(x, A)$ is a non-empty, closed and convex cone (a set A is a cone if for any $\alpha \geq 0$ and any $x \in A$, we have: $\alpha x \in A$).
2. If the set A is closed, prove that: $z \in \mathcal{N}(x, A)$ if and only if $\|z\|_2 = \mathrm{dist}(x + z, A)$.
3. If the set A is an epigraph of a convex function $f : \mathbb{R}^d \to \mathbb{R}$, i.e.,

$$\texttt{epi}\, f = \{(x, \gamma) \mid \gamma \geq f(x)\} \subseteq \mathbb{R}^{d+1},$$

   and the point $y = (x, f(x))$, then $z \in \mathcal{N}(y, A)$ if and only if $z = (\alpha g, -\alpha)$ where $g \in \partial f(x)$ and $\alpha \geq 0$.

Exercise 3 (Normal cones computing)**.** Compute the normal cones of the following sets:

1. $A = \mathbb{R}^d_+$ is the positive orthant, at the origin $x = 0$.
2. $A = \bar{\mathcal{B}}(0, 1)$ is the closed unit ball and $x \in \mathcal{B}(0, 1)$ in the interior of A and $x \in \bar{\mathcal{B}}(0, 1) \setminus \mathcal{B}(0, 1)$ at the boundary of A.

3. A set $A \in \mathbb{R}^{d+1}$ and $x = 0 \in \mathbb{R}^d$ where $A = \{(x, y) \in \mathbb{R}^{d+1} \mid y \geq \|x\|_1\}$.

Exercise 4 (Domain assumption necessity)**.** We already see in the lecture that for two convex functions $f_1, f_2 : \mathbb{R}^d \to \mathbb{R}$, $\partial(f_1 + f_2)(x) = \partial f_1(x) + \partial f_2(x)$. However, this is not necessarily true for *extended-valued* convex functions, i.e., $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$. Consider the indicator function of a convex set A, defined as:

$$\mathbf{1}_A(x) = \begin{cases} 0, & \text{if } x \in A \\ +\infty, & \text{otherwise} \end{cases}.$$

Consider A_1 and A_2 be two closed balls and $A_1 \cap A_2 = \{x\}$ is a singleton. Answer the following questions:

1. General question: Prove that for general convex functions f_1, f_2, we always have: $\partial f_1 + \partial f_2 \subseteq \partial(f_1 + f_2)$.
2. What is the subgradient of the functions $\mathbf{1}_{A_1}$ and $\mathbf{1}_{A_2}$ at x?
3. What is the function $\mathbf{1}_{A_1} + \mathbf{1}_{A_2}$? What is its set of subgradients at the point x? Conclude that $\partial \mathbf{1}_{A_1} + \partial \mathbf{1}_{A_2} \subsetneq \partial(\mathbf{1}_{A_1} + \mathbf{1}_{A_2})$.

Exercise 5 (Prove subgradient calculus under assumptions)**.** In this section, we will prove that for two convex functions $f_1, f_2 : \mathbb{R}^d \to \mathbb{R}$, we have:

$$\partial(f_1 + f_2)(x) = \partial f_1(x) + \partial f_2(x).$$

Answer the following questions:

1. Consider the sets:
$$\Omega_1 := \{(x, \lambda_1, \lambda_2) \mid \lambda_1 \geq f_1(x)\} \subseteq \mathbb{R}^{d+2},$$
$$\Omega_2 := \{(x, \lambda_1, \lambda_2) \mid \lambda_2 \geq f_2(x)\} \subseteq \mathbb{R}^{d+2}.$$
Prove that $\mathcal{N}((x, \lambda_1, \lambda_2), \Omega_k) = \mathcal{N}((x, \lambda_k), \Omega_k) \times \{0\}, k = 1, 2$.
2. Prove that $v \in \partial(f_1 + f_2)(x)$ if and only if $(v, -1, -1) \in \mathcal{N}((x, f_1(x), f_2(x)), \Omega_1 \cap \Omega_2)$.
3. Admit that $\mathcal{N}(z, \Omega_1 \cap \Omega_2) = \mathcal{N}(z, \Omega_1) + \mathcal{N}(z, \Omega_2)$ (warning: this is not always true, as we see in the previous exercise), conclude the proof.

Exercise 6 (Alternative projection as Polyak step-sizes)**.** Finding the interesetion of $\Omega_1 \cap \ldots \cap \Omega_n$ where $\Omega_k, k = 1, \ldots, n$ are non-empty, closed, convex sets, is an important computer science question. Assume that we can project onto each Ω_k efficiently, consider the following algorithm: at the ℓth iteration, we compute $\mathrm{dist}(x_\ell, \Omega_k), k = 1, \ldots, n$ and choose $p := \arg\max_k \mathrm{dist}(x_\ell, \Omega_k)$, then:

$$x_{\ell+1} = \mathrm{argmin}_{x \in \Omega_p} \mathrm{dist}(x, \Omega_p).$$

Answer the following questions:

1. Let $f(x) = \max(\mathrm{dist}(x, \Omega_1), \ldots, \mathrm{dist}(x, \Omega_n))$. Prove that f is convex, and 1-Lipschitz.
2. Prove that $\frac{x_{\ell+1} - x_\ell}{\|x_{\ell+1} - x_\ell\|}$ is a subgradient of f at x_ℓ. Hint: What is the subgradient of the distance function to a non-empty, closed and convex set?
3. Prove that $x_{\ell+1}$ is the update of the subgradient method for the function f with Polyak step-size $\Omega_1 \cap \ldots \cap \Omega_n \neq \emptyset$.
4. If $\Omega_1 \cap \ldots \cap \Omega_n \neq \emptyset$, what can we conclude? Can we find a point $x \in \Omega_1 \cap \ldots \cap \Omega_n$ at the end of the algorithm?